

5(a)	any four from: 1 (partial) geographical, isolation / separation / barrier; 2 little / no, interbreeding / gene flow (between puma populations) ; 3 different, environmental (conditions) / selection pressures ; 4 e.g. climate / vegetation / habitat / available prey ; 5 random / different, mutations ; 6 different, alleles selected for / gene pool / changes in allele frequency ; 7 allopatric (sub speciation) ;	4
5(b)	any three from: 1 obtain, blood / cells / tissue / DNA ; 2 from two (or more) individuals, at different locations / across range / from North and South America / from the different subspecies ; 3 use PCR to, replicate / amplify DNA ; 4 use, gel electrophoresis / DNA profiling / DNA fingerprinting ; 5 sequencing of DNA ; 6 use, genome / DNA, microarray ; R for gene expression 7 <i>ref. to</i> bioinformatics / database / (computer) software ; 8 compare similarity (of different sub species);	3

Question	Answer	Marks
5(c)(i)	any two from: <i>discontinuous (no mark) because:</i> 1 feature either present or absent ; 2 categoric ; 3 no range / no intermediates or qualitative data or does not show a normal distribution curve;	2
5(c)(ii)	any three from: 1 these features are controlled by, genes / alleles ; 2 <i>ref. to</i> genetic drift ; 3 (population went through a) <u>bottleneck</u> ; 4 low / reduced, number of alleles / genetic diversity / genetic variation / genetic polymorphism (in population) ; A some alleles lost 5 inbreeding ; 6 low / reduced, heterozygosity or high / increased, homozygosity ; 7 rare / deleterious, recessive alleles show their effects (as homozygous) ; 8 AVP ; e.g. number of breeding males reduced even more as some have, non-functional / undescended, testes	3

Question	Answer	Marks
5(c)(iii)	<i>any two from:</i> 1 Texas is closest to Florida ; ora A / B, distant / far away 2 (Texas and Florida pumas) are most closely-related / most genetically similar share most recent common ancestor ; ora A / B 3 Texas and Florida, climates / habitats / environment, are similar ; ora for A / B 4 AVP ; e.g. Texas has too many pumas	2